AMRO TABCHE

Ottawa, ON | +1 (343) 988-7403 | atabc081@uottawa.ca | linkedin.com/in/amro-tabche

TECHNICAL SKILLS

CAD / Simulation: SolidWorks, MATLAB/Simulink, FEA, structural optimization
BAS / Controls: Tridium Niagara, Alerton Compass, JACE 8000, BACnet, Modbus, HVAC, VAV, DDC, ASHRAE
Programming / Tools: Python, Arduino (C/C++), embedded systems, Microsoft Office, LaTeX, Git, Gantt Chart
Fabrication / Test: GD&T, ICT (bed-of-nails), 4-wire Kelvin, 3D Printing, MIG Welding, Sheet Metal
Languages: English, Arabic

WORK EXPERIENCE

Building Management Systems Engineering Training — **Aug 2023; Jul - Aug 2024**
Honeywell — **Riyadh, Saudi Arabia**

- Completed Honeywell-certified BAS training in Riyadh covering hands-on configuration of Alerton BACnet VAV HVAC controllers and Tridium JACE 8000 DDC systems, with BACnet IP/MSTP and Modbus protocol integration on supervised equipment at Sofitel Riyadh
- Trained on Alerton Compass programming for control logic and scheduling sequences; gained hands-on experience in Modbus RS-485 wiring, polling, and register mapping
- Practiced system integration and troubleshooting workflows across BAS architectures; produced structured inspection reports as part of certification deliverables

Mechanical Engineering Co-op — **Jan 2024 - Apr 2024**
Comtest Networks Inc. — **Nepean, ON**

- Supported manufacturing and prototyping workflows for telecom-sector PCB (printed circuit board) production, managing prototyping trials and process optimization initiatives across the full product lifecycle from design review through assembly
- Conducted in-circuit testing (ICT) of populated PCB assemblies using bed-of-nails test fixtures and 4-wire Kelvin resistance measurement at milliohm-level precision to detect shorting errors, opens, and continuity faults prior to downstream assembly
- Maintained and updated Bills of Materials (BOMs) and Material Resource Planning (MRP) data to support production scheduling and inventory accuracy
- Implemented ESD-safe handling procedures across work cells, improving safety compliance in an electronics manufacturing environment

PROJECTS

Compact Defensive Turret System — **Sep 2025 - Apr 2026**
Industry Capstone - ING Robotic Aviation Inc. — **Ottawa, ON**

- Designed a drone-countermeasure weapon platform integrating a 12-gauge shotgun (50-round magazine) into a two-axis pan/tilt mechanism achieving 360° continuous azimuth rotation and 90° elevation, constrained by strict size, mass, and power requirements
- Performed FEA-based structural analysis and optimization of three critical components (mounting plate, carriage plate damper interface, and guide rods) evaluated against structural strength, recoil response, mass, and aerodynamic drag criteria
- Designed a recoil management mechanism and all-weather sealed enclosure to maintain system reliability and structural integrity under operational loading conditions
- Developed a parametric optimization workflow directly linked to the SolidWorks CAD assembly, enabling automated geometry updates and real-time visual validation of optimal component configurations

One-Handed Pet Waste Scooping Device — **May 2025 - Jul 2025**
Accessible Design Category Design Day Winner — **Ottawa, ON**

- Designed and built a 380 g assistive device enabling a quadriplegic wheelchair user to retrieve dog waste single-handedly, integrating a servo-driven claw, Arduino-based control, rocker-switch input, magnetic bag attachment, and carbon-fiber shaft
- Conducted 7+ structured client testing sessions across grip force, reach range, bag removal time, and terrain adaptability; iterated on claw geometry and wrist strap sizing based on direct user feedback
- Delivered complete product documentation package including full user manual, BOM (~$45 prototype cost), and assembly instructions

Two-Wheel Voice-Controlled Drone (VC-CD) — **Oct 2023 - Dec 2023**

- Built an autonomous mechatronic two-wheeled voice-controlled platform on Arduino UNO with multi-modal behaviors: Grove speech recognition driving forward/reverse/turn/speed commands, HC-SR04 ultrasonic obstacle detection with automatic motor cutoff at <10 cm, and photoresistor-triggered LED headlight activation in low-light conditions; 8 W power draw with ~15 min runtime on a 4×AA battery pack
- Modeled three-channel sensor signal conditioning chain in Simulink (instrumentation amplification → second-order low-pass filtering → multiplexing → sample-and-hold → 12-bit ADC); designed full chassis in SolidWorks with mass-tuned lower compartment for stability under dynamic loading

EDUCATION & CERTIFICATIONS

University of Ottawa - BASc, Mechanical Engineering · May 2026
Honeywell-certified: Tridium Niagara 4.13 (Networking, JACE 8000, BACnet/Modbus); Alerton (VLC, BACnet IP/MSTP, Modbus, Compass)